Professional Program in

Data Science, Machine Learning, AI & GenAI

eict IIT Roorkee

E&ICT, IIT Roorkee

The IoT Academy
Connecting The Unconnected

Projects

Duration: 10 Months | 240 Hours

DS | ML | DL | AI | Generative AI

Advanced Gen AI, Agentic AI & LLMs Are Covered



About The IIT Roorkee

Indian Institute of Technology Roorkee (IIT Roorkee), one of the oldest technical institutions in Asia, was established in **1847** as the Thomason College of Civil Engineering and was later converted to an IIT in 2001

The institute offers a wide range of academic programmes including BTech, BArch, BS, MTech, MSc, MBA, MDes, and PhD across its 23 academic departments, centres, and schools covering engineering, sciences, architecture, management, humanities, and social sciences.

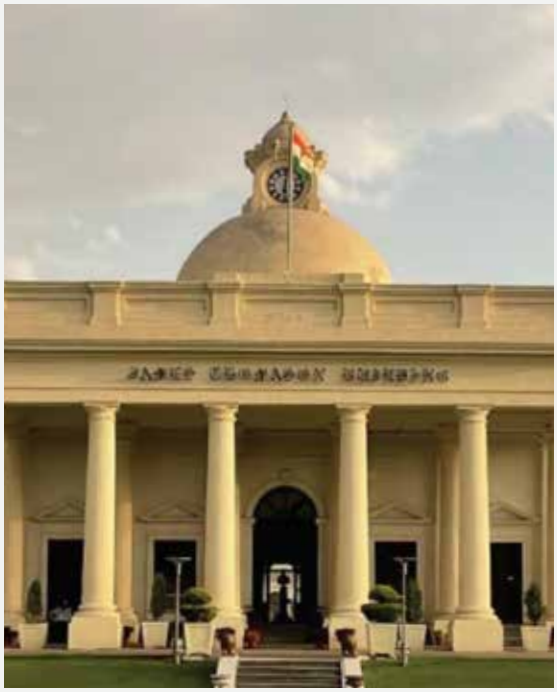
Over the years, **IIT Roorkee** has developed state-of-the-art infrastructure, a strong research ecosystem, and a rich legacy of excellence in technical education and innovation. The institute continues to be recognized nationally and internationally for its academic rigor, research output, and contributions to industry and society.



About The E&ICT Academy IIT Roorkee

Electronics & ICT Academy, IIT Roorkee aims to strengthen the skills of faculty members, working professionals, and students from Engineering, Science, Commerce, and Polytechnic institutions through specialized short-term training programs.
The Academy focuses on both fundamental and advanced areas of Information Technology, Electronics & Communication, Product Design, and Manufacturing.

In addition, E&ICT Academy, IIT Roorkee organizes customized industry-oriented training programs, faculty development programs (FDPs), and research promotion workshops to bridge the gap between academia and industry.
These initiatives are designed to enhance technical competency, encourage innovation, and promote collaborative research in emerging technologies.



About The IoT Academy

The IoT Academy, established in 2017 is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. E&ICT Academy, IIT-Roorkee, IIT-Guwahati and IIT-Kanpur for Advance Certification courses to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

30,000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes





About The Program

Most of the modern technological innovations you see today are powered by Data Science, Machine Learning, AI, and Generative AI. This 10-months Professional Program In Data Science, Machine Learning, AI & GenAI is thoughtfully designed by E&ICT Academy, IIT Roorkee, in collaboration with industry experts, to help you master the exact skills the market demands. The primary goal of this certification program is to prepare you for the most in-demand roles in Data Science, Machine Learning, AI, and Generative AI. You will gain expertise in Artificial Intelligence & Data Technologies, Advanced Analytics, Intelligent Systems, Machine Intelligence, Prompt Engineering and AI/ML-driven Solutions through multiple real-world projects in Machine Learning, Deep Learning, and Generative AI — transforming you into a highly skilled and industry-ready New Age DS, ML, AI & Gen AI Specialist.

Integrated With Gen AI, Agentic AI & LLMs

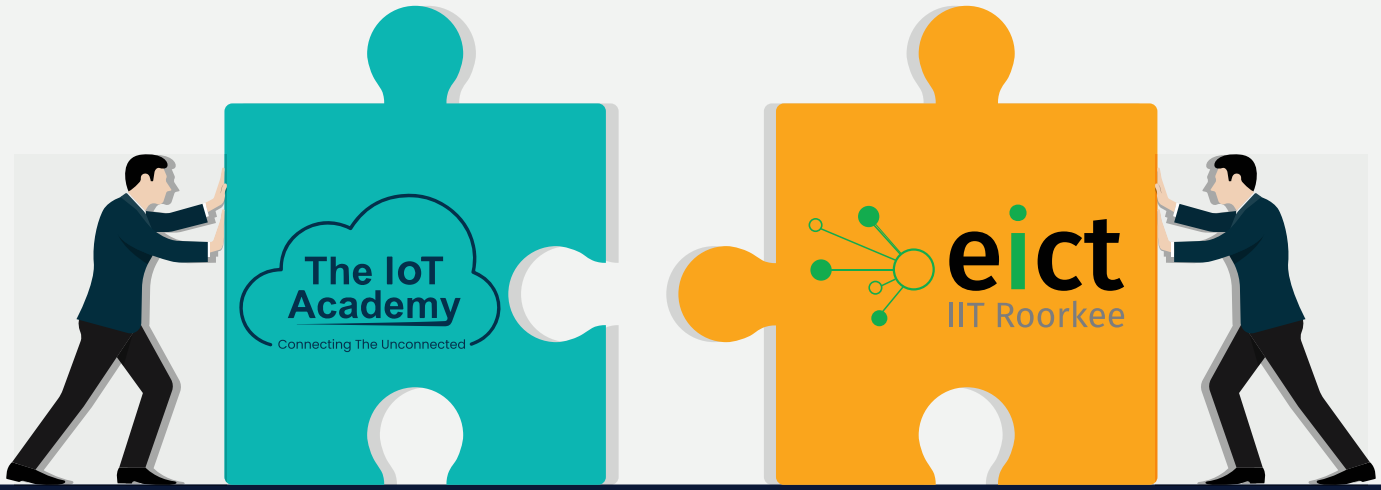
- Gen AI & Agentic AI Modules & Architecture
- Building LLM Applications
- Gen AI & Agentic AI Tools & Case Studies

PARTNERSHIP



About The Partnership

Professional Program In Data Science, Machine Learning, AI & GenAI is offered jointly by The IoT Academy and E&ICT Academy, IIT Roorkee to prepare the future workforce in Machine Learning and AI



Capstone Projects

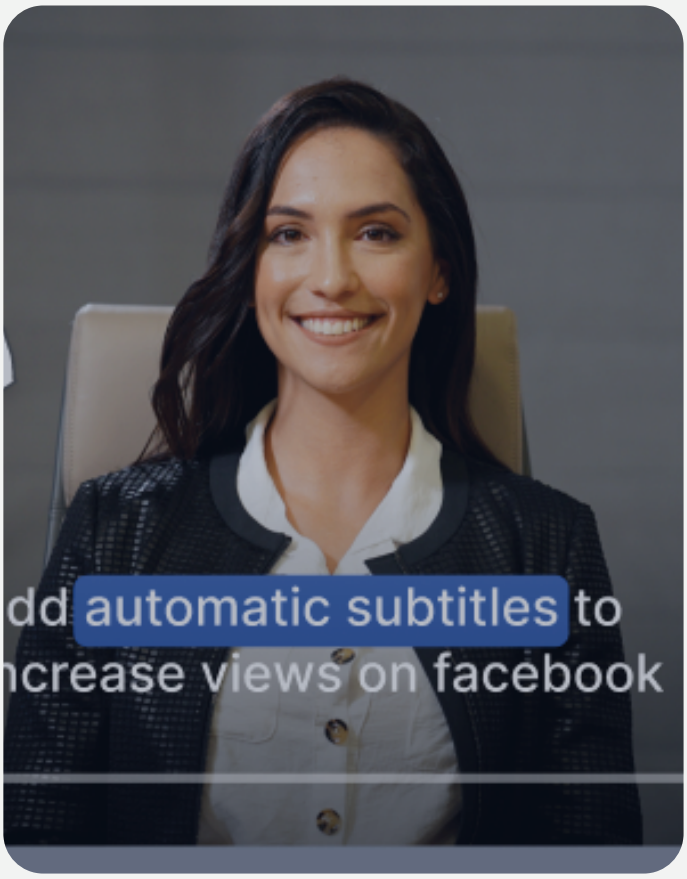
Industry projects will be a part of your Professional Program In Data Science, Machine Learning, AI & GenAI to consolidate your learning. Industry projects will ensure you have the real-world experience to start your career in DS, ML-AI.

- **10+ Essential Tools**
- **Designed by Industry Experts**
- **Get Real-world Experience**

Auto Image Captioning

Automatic image captioning is widely used by search engines to retrieve and show relevant search results to the users. For example - to categorize personal multimedia collections, for automatic product tagging in online catalogs, and other areas of business and researches. Use CNN and LSTM to create a model that can automatically add captions to the image.

Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib





Big Mart - Customer Segmentation

The data scientists at Big Mart have collected 2013 sales data of 1559 products across 10 stores in different cities. Big Mart CEO wants to understand the customer demographics and customer retention (Customer who can converge easily) so that the marketing team can market their products and services by conducting various strategies accordingly.

Tools you will be using: Pandas, scipy, Numpy



Building a RASA Based Chatbot

Rasa is a framework for developing AI-powered, industrial grade, powerful chatbots. The developers use it to create intent-based chatbots. In this project, we are going to understand some of the most importantbasic aspects of the Rasa framework and use RASA NLU and RASA CORE to build a conversational chatbot.

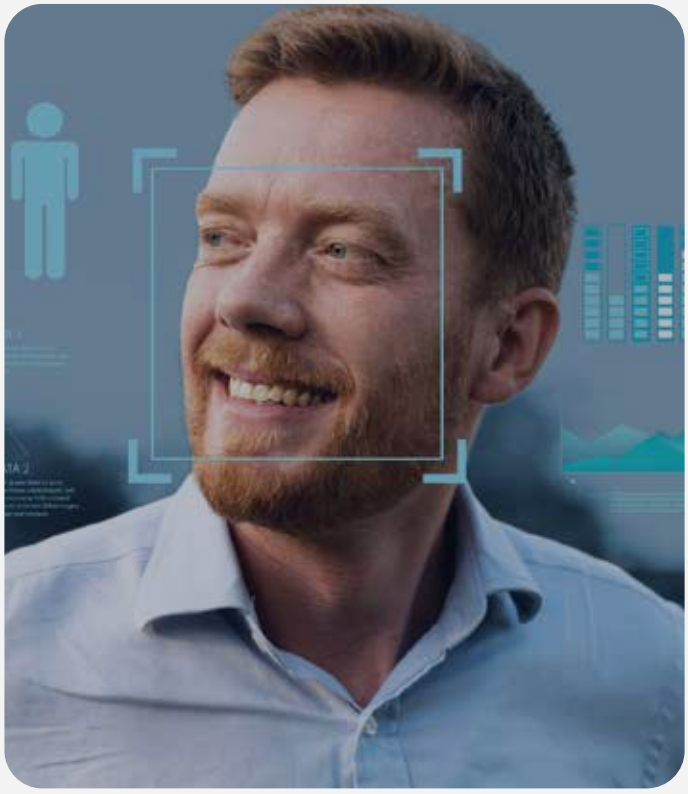
Tools you will be using: RASA NLU, RASA Core



Real-Time Age, Emotion and Gender Detection using CNN

Use CNN and OpenCV to create a model which would detect the person's age, emotion, and age in real-time.

Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib



Telecom Churn Prediction

A telecom company wants you to analyze its data, to keep its customers. You will be provided with the 'Telecom Churn' dataset. Use it to create a model to predict which customer will switch to other telecom service providers, based on the relevant customer data.

Tools you will be using: Pandas, Numpy,, Seaborn, Matplotlib, SKLearn



Smart Surveillance System

A shopping centre needs a surveillance system to detect persons and other items. As a machine learning engineer, you will create a model to detect objects using a pre-trained Mask RCNN model.You will be using OpenCV, Tensorflow2, Keras, Numpy, Pandas, Matplotliband others.

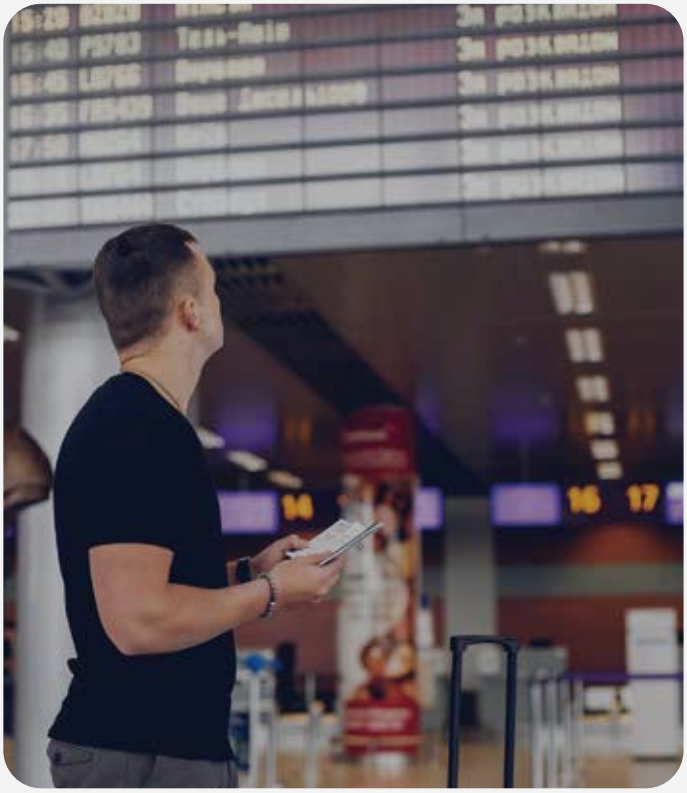
Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib



Forecast Air Passenger Traffic

An Airline called Star Air has the data of its passengers across months. The data is classified in date/time and the passengers travelling per month. Build a model to forecast the demand (passenger traffic) in Airplanes. You will learn to use Pandas, Scipy, Numpy with hands-on experience of other tools, features and libraries.

Tools you will be using: Pandas, scipy, Numpy



Housing Price Prediction

The dataset is collected from the 1990 California census containing data of one row per census group. The dataset has various demographics and details captured. Based on this data, we have to create a model using Pandas, Scipy, Numpy that can determine the housing price of the house based on the details provided.

Tools you will be using: Pandas, scipy, Numpy



Traffic Sign Classification using CNN in Tensorflow 2.0

Detection and recognition of traffic signs are crucial for the development of self- driving cars, which have a direct impact on driving behaviors. You will learn to build a CNN model using OpenCV, TensorFlow2, Keras, Numpy, Pandas, Matplotlib to detect and classify the traffic signals for new self-driving cars.

Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib



Analyze & Visualize Employee Attrition

Employee retention is one of the biggest metrics that a company should have in mind when thinking of growth. Employee attrition is caused when the total strength of the company is greatly reduced as more employees leave the company than expected. Uncover the factors that lead to employee attrition and explore the reasons as to why people are leaving the organization and predict whether an employee will leave the organization or not by creating a Web App using Streamlit that takes inputs from user's online.

Tools you will be using: Sklearn, PyCaret, Streamlit, SHAP, Pandas-Profiling



Analyze & Visualize Video Games Sales

SRS Ltd. is a Korean and video game company headquartered in Seoul. You as a Data Scientist is requiredto analyze the trend in Global Sales according to the Genres on the Video Games Sales Dataset from 1980 & visualize the change in Net Sales of different publishers from the year 2005 to 2015.

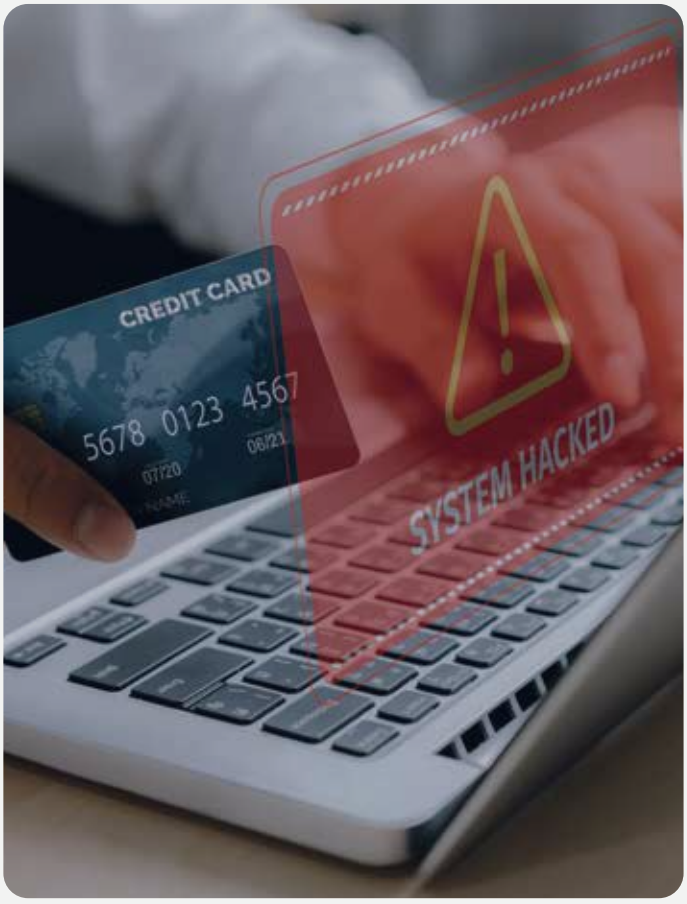
Tools you will be using: Plotly Express, Streamlit



Identify fraudulent credit card transactions

Credit card fraud happens when a fraudster or a thief steals your credit card or the information from that card to make unauthorized purchases in your name or take out cash advances using your account. Credit card companies such as Citibank, HSBC, and American Express need to recognize fraudulent credit card transactionsso that customersare not charged for items that they did not purchase.

Tools you will be using: Sklearn, PyCaret, Streamlit, ELI5, Pandas-Profiling



Climate Impact Analysis

Description: Use generative AI to analyze and visualize the effects of climate change through image and text data synthesis.

Tools and Technologies: Diffusion Models, GANs, TensorFlow, PyTorch.

Outcomes: Learn to work with multi-modal data and generate insights using generative AI.



Text-to-Image Generation

Description: Combine the power of NLP and CV to generate realistic images from textual descriptions, such as creating visual prototypes for design concepts or generating fictional landscapes.

Tools and Technologies: PyTorch, TensorFlow, OpenCV, Diffusion Models, GANs, Transformers.

Outcomes: Learn to preprocess text and image data, design transformer-based models, and implement diffusion models for image synthesis.





Interactive Chatbot for Healthcare

Description: Build a conversational agent using GPT-based LLMs that assists users with healthcare queries, such as symptom checks or medication reminders.

Tools and Technologies: Rasa, PyTorch, Hugging Face Transformers, Streamlit, NLP Libraries (spaCy, NLTK).

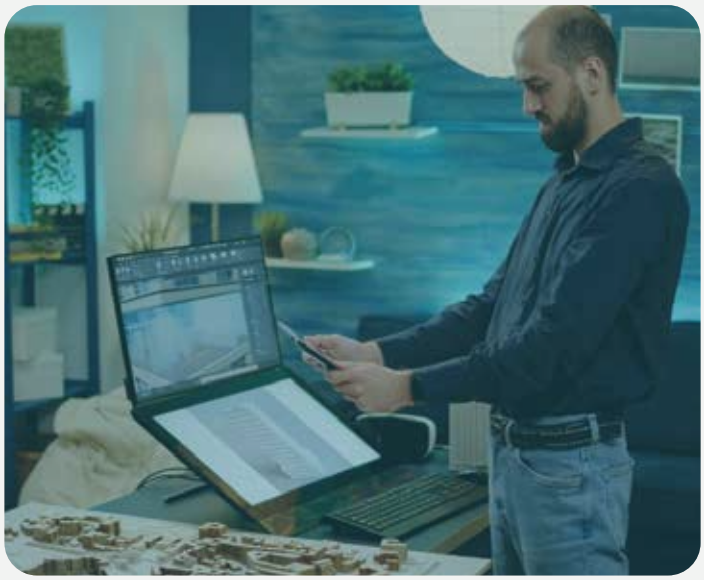
Outcomes: Gain experience fine-tuning LLMs for domain-specific tasks, handling conversational context, and deploying chatbots.

Generative Art Creation

Description: Develop a system to generate artwork using GANs or style transfer techniques. The model will create unique artistic styles or merge different styles into new creations.

Tools and Technologies: TensorFlow, PyTorch, StyleGAN, OpenCV.

Outcomes: Understand GAN architecture, apply advanced loss functions, and explore applications in digital art and design.





Predictive Maintenance with Generative Models

Description: Use Variational Autoencoders (VAEs) to detect anomalies in industrial sensor data and predict maintenance needs.

Tools and Technologies: PyTorch, TensorFlow, VAE, Pandas, NumPy.

Outcomes: Learn latent space representation, anomaly detection, and how generative AI can enhance predictive maintenance.

AI-Powered Content Generation for Marketing

Description: Create a generative AI model to assist marketing teams in generating product descriptions, blog posts, and ad copies.

Tools and Technologies: GPT, Hugging Face, PyTorch, TensorFlow.

Outcomes: Work with fine-tuning strategies for LLMs, data cleaning, and evaluating text quality using BLEU and ROUGE metrics.





Video Synthesis for Personalized Education

Description: Build a system that uses GANs to create educational video content tailored to individual learning styles and topics.

Tools and Technologies: GANs, TensorFlow, PyTorch, FFMPEG for video processing.

Outcomes: Understand video generation techniques, data preprocessing for videos, and model evaluation.

Voice Cloning and Audio Synthesis

Description: Develop a model to synthesize human-like speech or clone specific voices using generative models like WaveNet or Tacotron.

Tools and Technologies: TensorFlow, PyTorch, Librosa, WaveNet.

Outcomes: Learn to preprocess audio data, generate waveforms, and build real-time voice applications.





Smart Surveillance System Using Generative Models

Description: Detect and track anomalies or objects in video footage using generative models for data augmentation and anomaly detection.

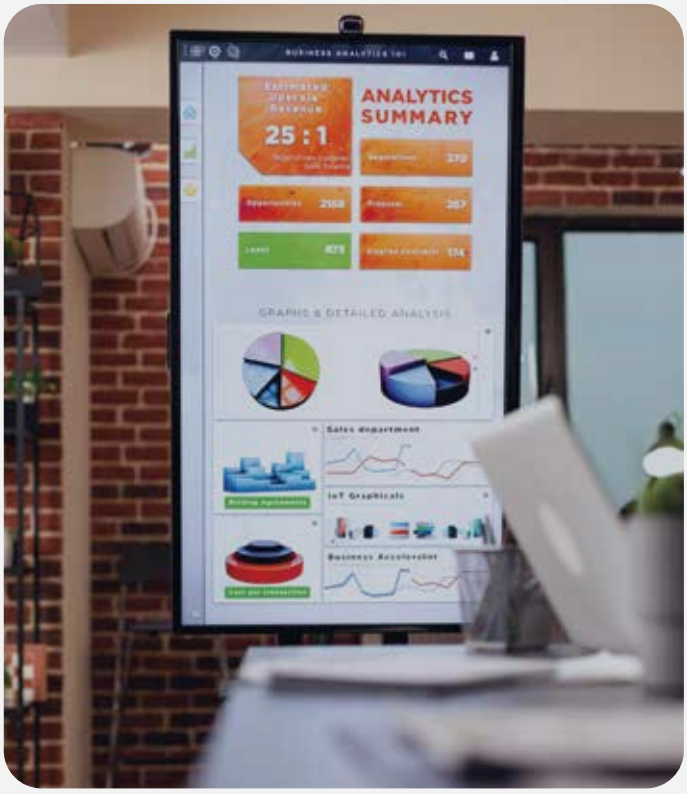
Tools and Technologies: TensorFlow, PyTorch, OpenCV, GANs.

Outcomes: Explore applications in computer vision, object detection, and real-time surveillance.

Customer Growth Intelligence Dashboard

This project focuses on analyzing customer acquisition, retention, churn, and engagement data to identify growth trends and support data-driven business strategies.

Tools you will be using:
Python, Pandas, SQL, Power BI / Tableau



Startup Financial Health Radar

This project analyzes key startup financial metrics such as revenue, burn rate, runway, and profitability to evaluate overall financial stability and performance.

Tools you will be using:
Python, Pandas, Excel, SQL



People Analytics: Silent Attrition Signals Explorer

This project examines workforce data to detect early signs of silent attrition by analyzing engagement, performance, and behavioral patterns.

Tools you will be using:
Python, Pandas, Scikit-learn, Power BI / Tableau



Viral Content Intelligence System

This project analyzes social media content and audience engagement data to identify patterns that contribute to high-performing and viral content.

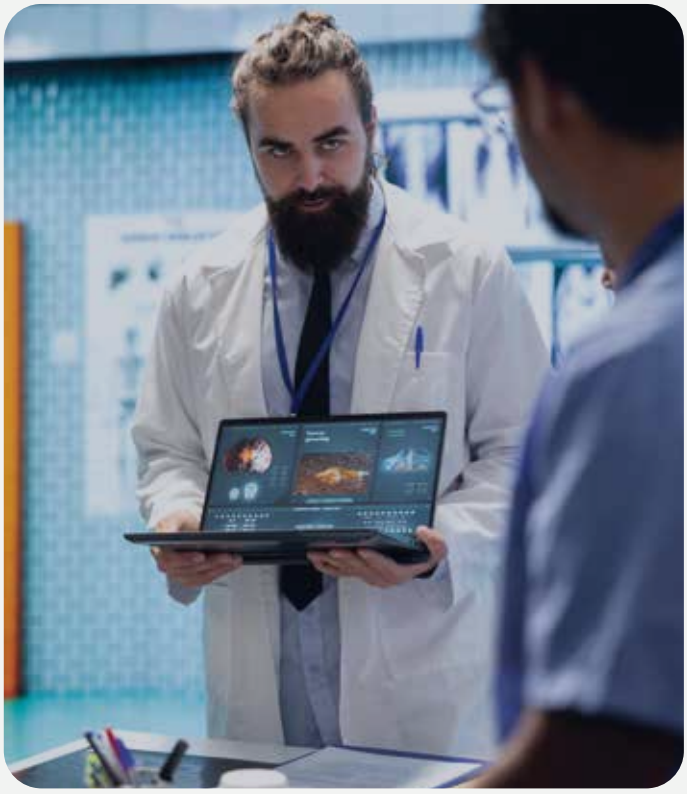
Tools you will be using:
Python, Social Media APIs, NLP Libraries



Public Health Intelligence & Policy Insight Lab

This project uses public health datasets to analyze trends, identify risk factors, and generate insights for informed policy and planning decisions.

Tools you will be using:
Python, Pandas, Data Visualization Tools



Growth Experimentation & A/B Intelligence Lab

This project focuses on designing and analyzing A/B experiments to evaluate growth strategies, user behavior, and conversion performance for data-driven optimization.

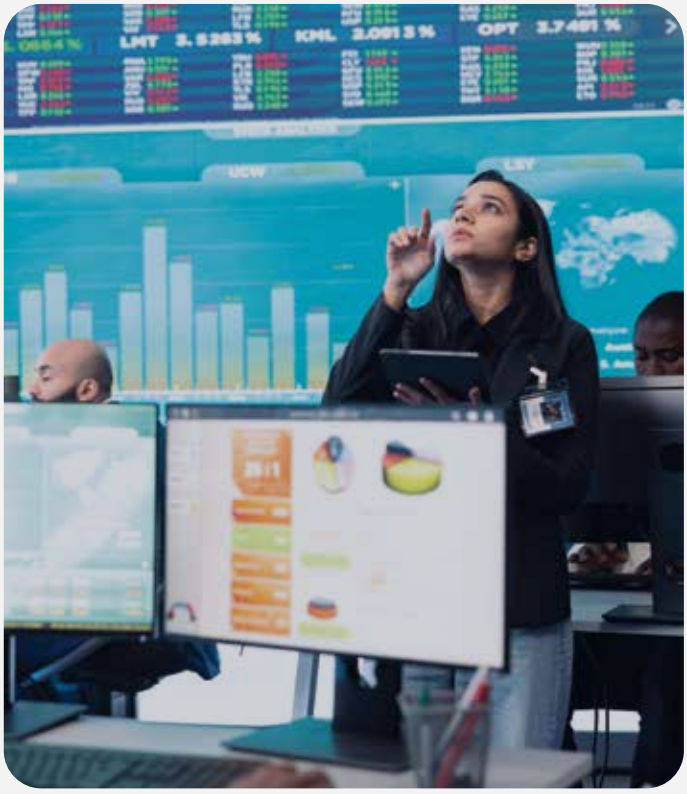
Tools you will be using:
Python, Pandas, SQL, A/B Testing Frameworks, Power BI / Tableau



Quant Risk & Market Volatility Intelligence System

This project analyzes market data to measure financial risk, volatility, and price fluctuations, supporting quantitative and risk-aware investment decisions.

Tools you will be using:
Python, Pandas, NumPy, Time Series Analysis, Financial Analytics Libraries



Fair Pay, Bias & Workforce Ethics Analyzer

This project analyzes compensation and workforce data to detect pay gaps, bias patterns, and ethical risks, promoting fairness and transparency in organizations.

Tools you will be using:
Python, Pandas, Statistical Analysis, Data Visualization Tools



Customer Churn Early-Warning Engine

This project focuses on predicting customer churn by analyzing behavioral and transactional data to enable early intervention and retention strategies.

Tools you will be using:
Python, Pandas, Scikit-learn, SQL, Machine Learning Models



AI-Driven Credit Risk Scoring System

This project uses machine learning techniques to assess creditworthiness by analyzing customer financial and behavioral data to minimize lending risk.

Tools you will be using:
Python, Pandas, NumPy, Scikit-learn, Financial Risk Models



Behavioral & Personality Intelligence Model

This project analyzes user behavior data to model personality traits and behavioral patterns for enhanced personalization and decision-making insights.

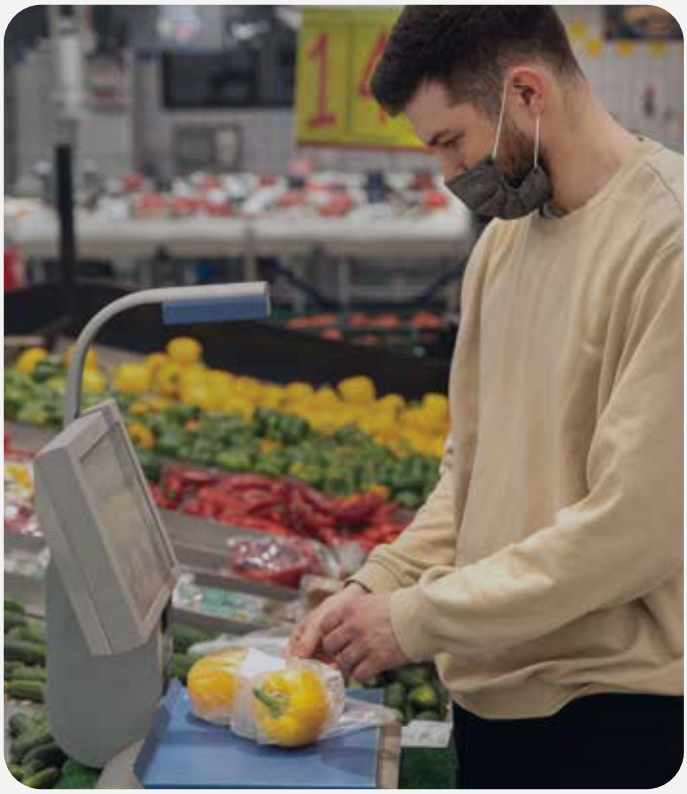
Tools you will be using:
Python, Pandas, NLP Libraries, Machine Learning Techniques



Retail Cross-Sell Recommendation Engine

This project builds a recommendation system to identify cross-sell opportunities by analyzing customer purchase history and product relationships.

Tools you will be using:
Python, Pandas, Recommendation Algorithms, Collaborative Filtering



Socioeconomic Segmentation & Inequality Mapping System

This project analyzes socioeconomic data to segment populations and visualize inequality patterns across regions for data-driven policy insights.

Tools you will be using:
Python, Pandas, GIS Tools, Data Visualization Libraries



Multimodal Sentiment & Emotion Intelligence System

This project analyzes text, images, and videos from social media to detect sentiments and emotions, helping understand public opinion and user behavior.

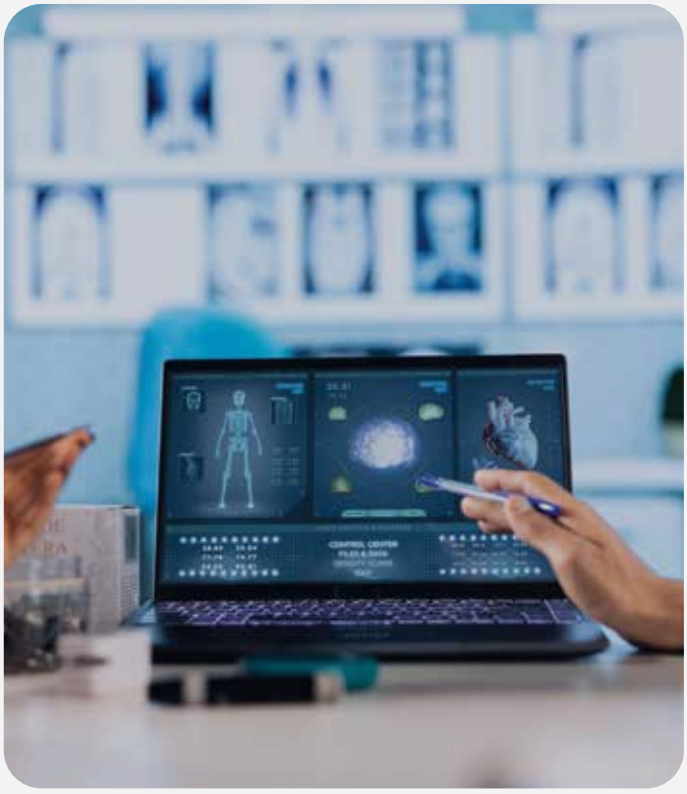
Tools you will be using:
Python, NLP Libraries, Computer Vision Libraries, Deep Learning Frameworks



AI-Assisted Medical Imaging Diagnosis System

This project uses medical imaging data to assist in disease detection and diagnosis by identifying patterns and anomalies in medical images.

Tools you will be using:
Python, OpenCV, Deep Learning Frameworks, Medical Imaging Libraries



Legal Outcome Prediction & Decision Support AI (Law)

This project analyzes legal case data to predict possible outcomes and provide decision-support insights for legal professionals.

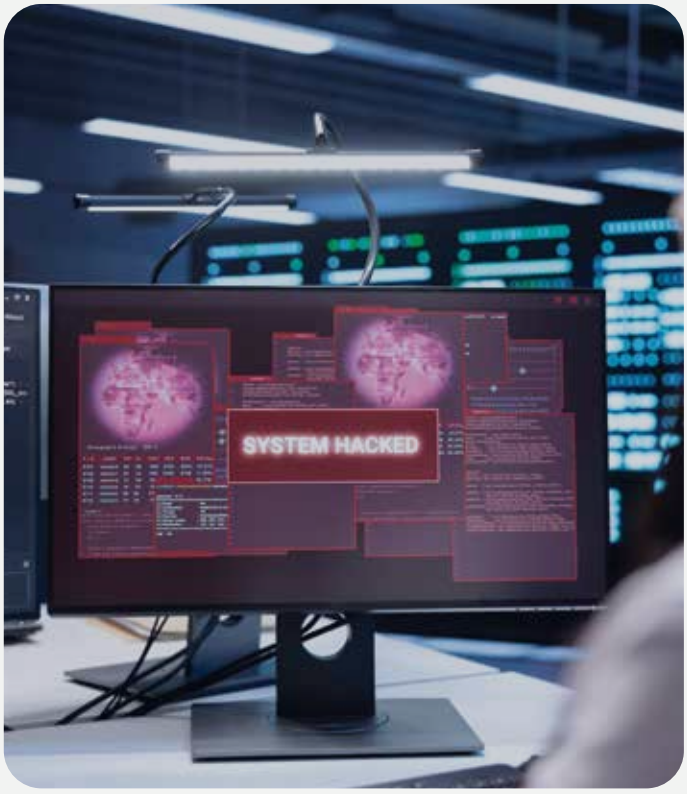
Tools you will be using:
Python, NLP Libraries, Machine Learning Models, Legal Data Sources



AI-Powered Threat Detection & Surveillance Intelligence

This project focuses on detecting potential threats by analyzing surveillance data, enabling proactive security monitoring and risk assessment.

Tools you will be using:
Python, Computer Vision, Deep Learning Models, Surveillance Analytics Tools



Industrial Failure Prediction & Reliability AI System

This project predicts equipment failures by analyzing sensor and operational data to improve system reliability and reduce downtime.

Tools you will be using:
Python, Pandas, Machine Learning Models, Time Series Analysis



Autonomous Marketing Campaign AI Agent

This project builds an autonomous AI agent that plans, executes, and optimizes marketing campaigns by analyzing audience behavior and campaign performance.

Tools you will be using:
Python, GenAI Frameworks, Marketing APIs, Data Analytics Tools



Explainable GenAI Financial Copilot (RAG-Based)

This project develops a GenAI-powered financial assistant using Retrieval-Augmented Generation to provide explainable insights from financial data and reports.

Tools you will be using:
Python, LLMs, RAG Frameworks, Vector Databases, Financial Datasets



Agentic AI Hiring & Interview Automation System

This project automates hiring workflows by screening resumes, scheduling interviews, and generating interview insights using agent-based AI systems.

Tools you will be using:
Python, NLP Libraries, Agentic AI Frameworks, HR Analytics Tools



AI-Driven Drug Discovery Intelligence Agent

This project applies AI techniques to analyze molecular and biomedical data to accelerate drug discovery and candidate identification.

Tools you will be using:
Python, Deep Learning Frameworks, Bioinformatics Tools, Molecular Data Libraries



Multimodal Clinical Documentation & Summarization AI

This project uses multimodal data such as text and voice to generate accurate clinical documentation and automated medical summaries.

Tools you will be using:
Python, NLP Libraries, Speech-to-Text Models, Healthcare Data Tools



GenAI Legal Research & Drafting Copilot

This project builds a GenAI-powered copilot to assist in legal research, case analysis, and automated drafting of legal documents.

Tools you will be using:
Python, LLMs, NLP Libraries, Legal Knowledge Bases



Tools

jupyter | tableau | Midjourney | ChatGPT | DALL·E
Gemini | Google Cloud AutoML Vision | Amazon SageMaker | GitHub | kaggle
Power BI | perplexity | deepseek | git | OpenAI

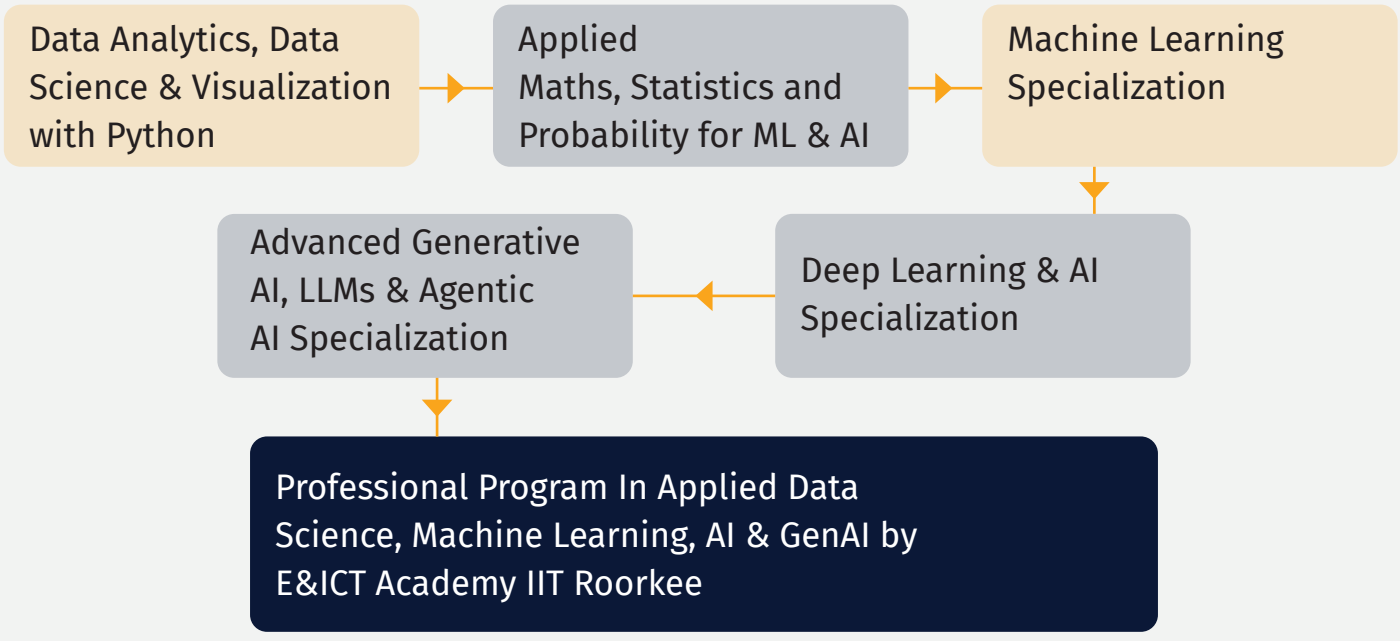
Languages

python | SQL | R

Frameworks

pandas | bokeh | matplotlib | OpenCV | spaCy
NumPy | scikit-learn | PyTorch | TensorFlow | MySQL
Keras | Hugging Face | TensorFlow Lite

Learning Path



E&ICT, IIT Roorkee

Professional Program In
Data Science, Machine Learning AI & GenAI

By E&ICT Academy, IIT Roorkee

WhatsApp/Call

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